

Advanced Theories of Bonding

Chapter 5

Chemistry: Atoms First 2nd Ed.

Louis Pascal Riel

UConn Avery Point

Fall 2025

1127 Housekeeping Oct 9th Exam II material

- Quiz 4 – Done
- Exam II – Thurs Oct 16th – covers Quiz 3 & 4 and today.
- HW see due dates →
 - Aktiv Quiz Daily Feedback
 - MIDTERM GRADES
- Extra Credit NACE Journal 3 & Exam II
Extra Credit Discussion Board (Monday by 4:00 pm)

A6: Chapter 4 part 2: Lewis Structures & Formal Charges

Homework

Start Date: Monday, Sep 29, 2025 1:28 PM

Due Date: Tuesday, Oct 14, 2025 11:59 PM

100 problems

DUE IN 7 DAYS

A7: Chapter 4 part 3: electron pair/molecular Geometries, Bond Angles, Polar Molecules, Intermolecular Forces, Solubility

Homework

Start Date: Monday, Sep 29, 2025 1:32 PM

Due Date: Thursday, Oct 16, 2025 1:59 AM

70 problems

DUE IN 13 DAYS

A8: Chapter 5 Advanced Theories of Bonding

Homework

Start Date: Monday, Feb 26, 2024 1:38 PM

Due Date: Tuesday, Oct 21, 2025 11:59 PM

40 problems

Lewis Structures and VSEPR are great!

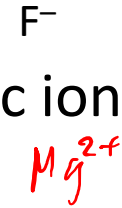
↳ Geometries are "Shapes"

- They predict
 - Covalent bonds
 - Molecular geometries
- But they can't reconcile or predict the following:
 1. s & p & d atomic orbitals versus the observed bond angles like 120° or 109.5°
 2. Magnetic behavior of O_2 vs magnetic inertness of N_2
- Requires further elaborate systems to describe these properties (1 & 2)

Atomic Orbital Overlap vs. Molecules

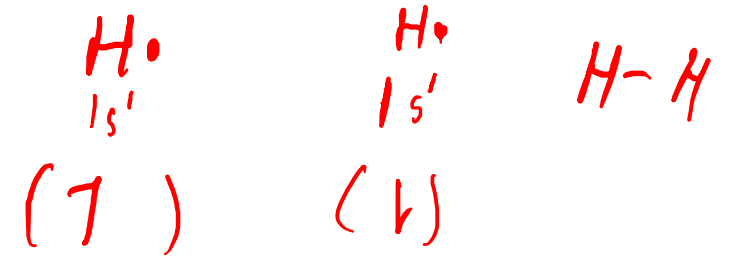
- Recall: **atomic orbitals** describe probable locations of electrons around an atom.

- These orbitals are for a single **atom** or monoatomic ion, not one in a molecule.



- To describe electrons in molecules, multiple theories may be used.
- Valence bond theory**
- Hybridization**
- Molecular orbital theory**

Valence Bond Theory (VBT)

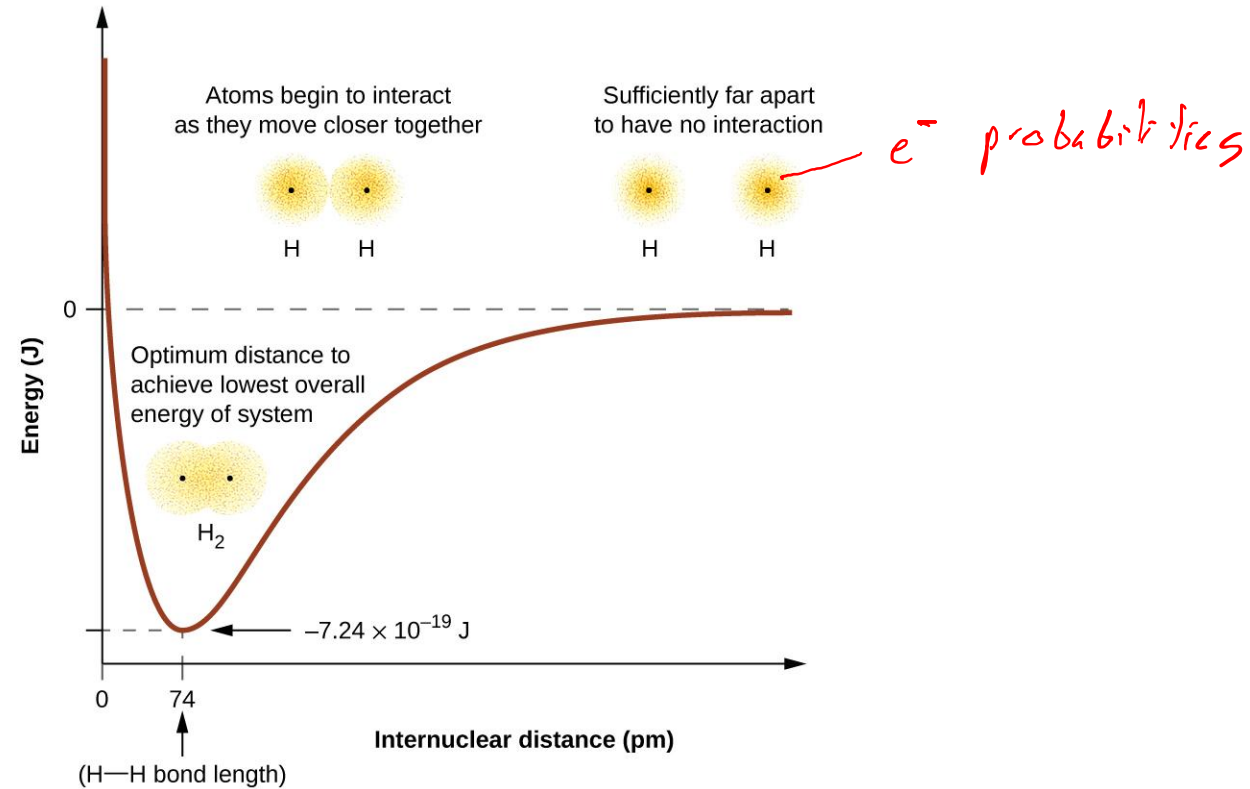


- VBT describes a covalent bond as the **overlap** of two half-filled atomic orbitals.
 - Each atom has an orbital that contributes a single electron to the bond.
- 2 attributes are needed for a covalent bond:
 - 2 orbitals must overlap.
 - Single electrons combine to form an electron pair.
- The greater the overlap between orbitals, the stronger the bond.

↳ 2 pieces Duct tape

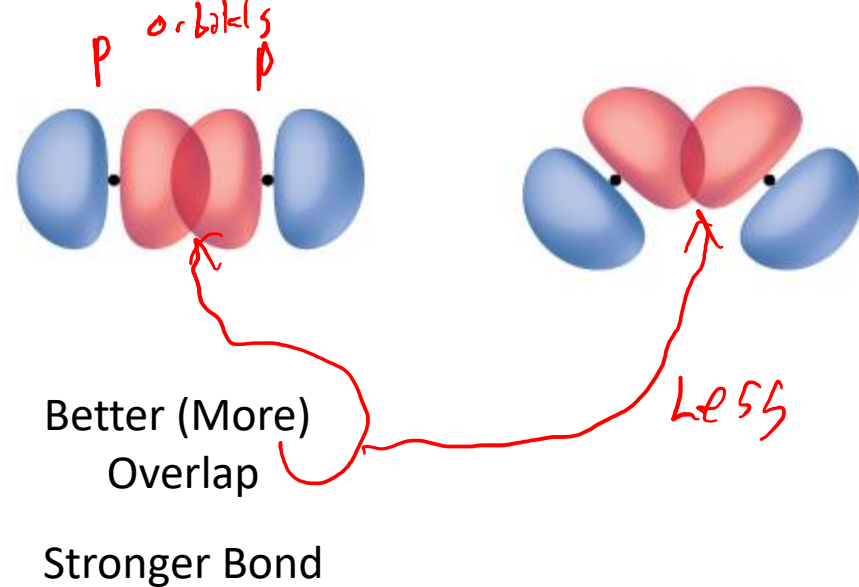
Valence Bond Theory (VBT)

Lower Energy = Greater Stability



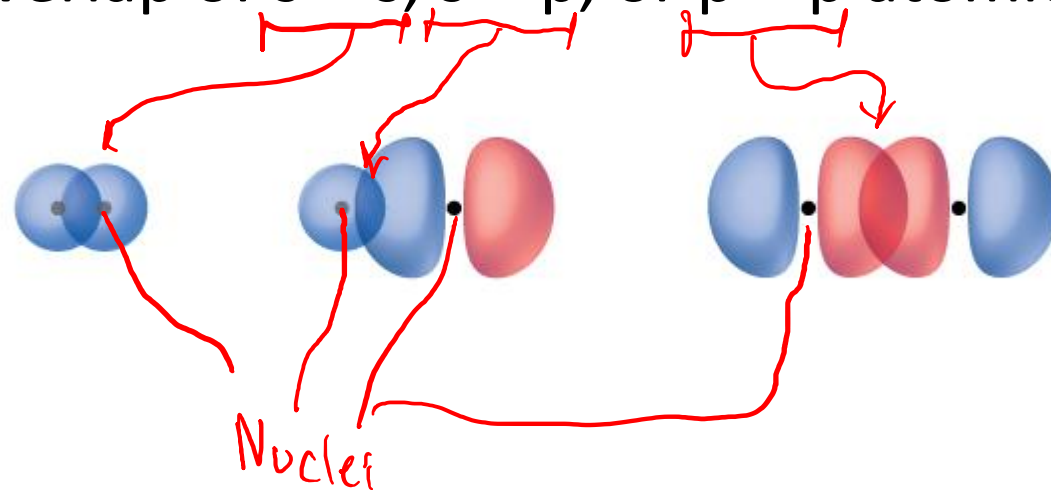
The greater the overlap between orbitals, the stronger the bond.

- Orientation matters too!
 - Greater overlap is possible when orbitals overlap **directly between the two nuclei**.
 - The bold above can also be described as an **internuclear axis**



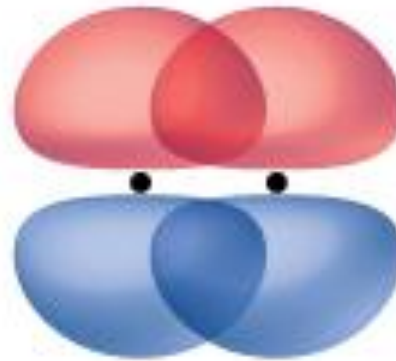
Sigma (σ) Bonds

- A σ bond is a **covalent bond** in which the electron density is concentrated in the region **along the internuclear axis**.
 - In VBT's description of Lewis structures, **single bonds are σ bonds**.
- Created from overlap of s – s, s – p, or p – p atomic orbitals.



Pi (π) Bonds

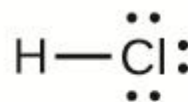
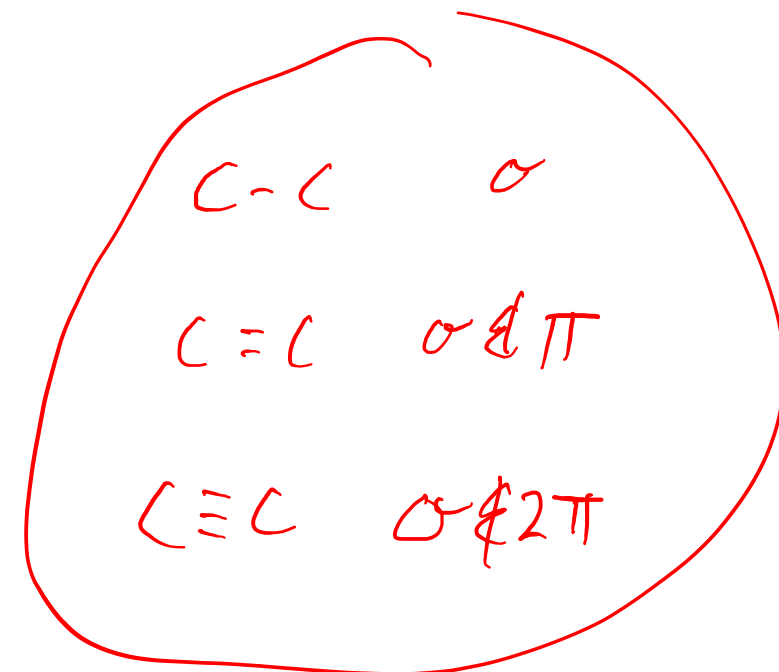
- A π bond is a covalent bond created from **side-by-side overlap** of two p orbitals.
 - Regions of orbital overlap lie on opposite sides of internuclear axis.
 - Along the axis, there is a **node**, where the electrons **cannot be located**.



Multiple bonds

single bond = σ bond.

- A double bond is composed of 1 σ and 1 π bond
 - $C=C$
- A triple bond is composed of 1 σ and 2 π bond
 - $C\equiv C$



One σ bond
No π bonds



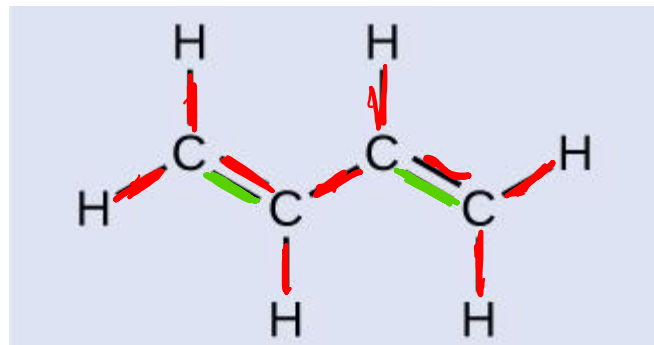
One σ bond
One π bond



One σ bond
Two π bonds

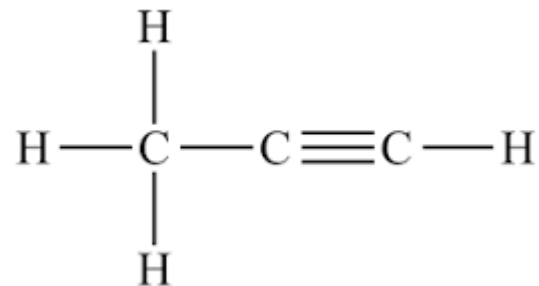
Practice:

- How many σ and π bonds are in butadiene, a monomer used to make synthetic rubber?



ICE Sigma Bonds

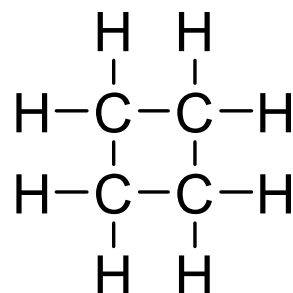
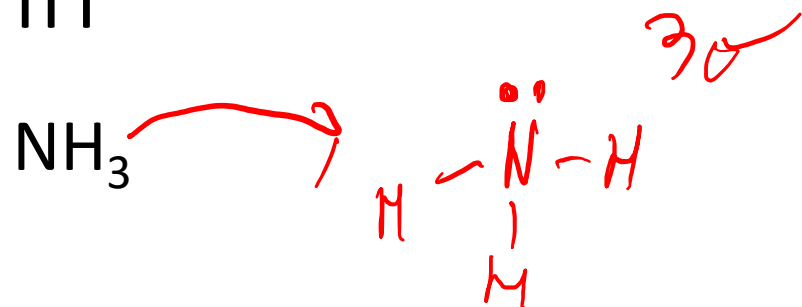
- Note the Lewis structure here →
 - 5 s



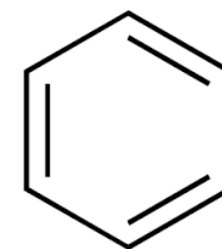
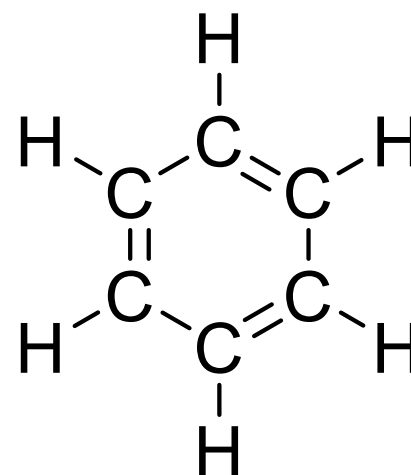
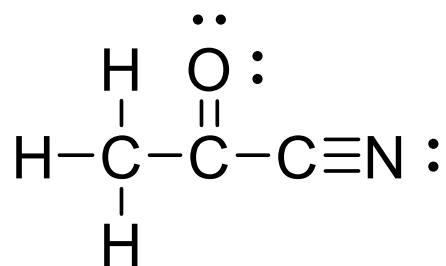
1. How many sigma (σ) bonds are present in the molecule **Propyne**, whose condensed structural formula is $\text{CH}_3\text{C}\equiv\text{CH}$?

- A. 5
- B. 6
- C. 7
- D. 8

How many σ bonds and π bonds are present in



These are equivalent to the structures to the right



ICE Double Bonds– 30 s

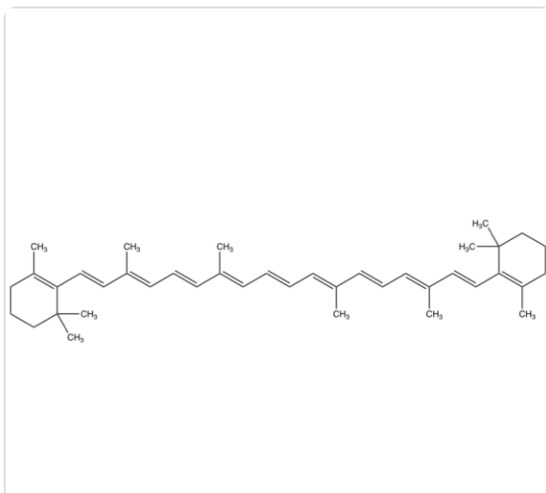
9:45 AM Thu Feb 23

92%



Show Answer

β -carotene is an antioxidant and a precursor to vitamin A found in carrots and other brightly colored foods. How many double bonds are present in the structure of β -carotene shown?

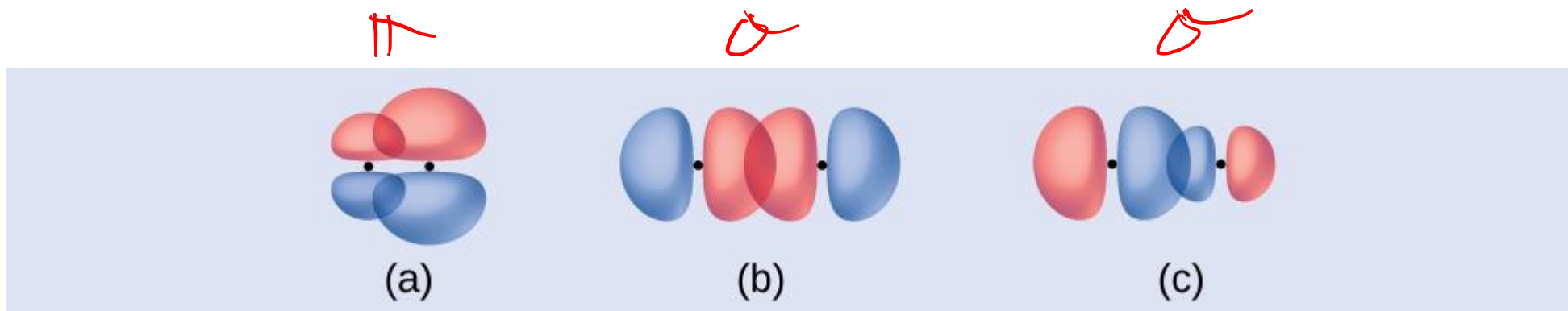


1	2	3	✕
4	5	6	C
7	8	9	
+/-	.	0	x 10

Practice:

Identify each illustration as depicting a σ or π bond:

- (a) side-by-side overlap of a $4p$ and a $2p$ orbital
- (b) end-to-end overlap of a $4p$ and $4p$ orbital
- (c) end-to-end overlap of a $4p$ and a $2p$ orbital



Bond Dipole Moments – Not Tested on

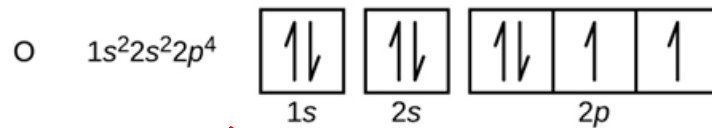
- The dipole moment of a bond is defined as charge times distance.
- Unit for bond dipole moment is the **Debye (D)** = $3.336 \times 10^{-30} \text{ C m}$
 - C m = Coulomb meters (charge x distance)
- HCl has a 1.08 D bond dipole
 - More polar bond than below
- HI has a 0.44 D bond dipole

Hybrid Orbitals

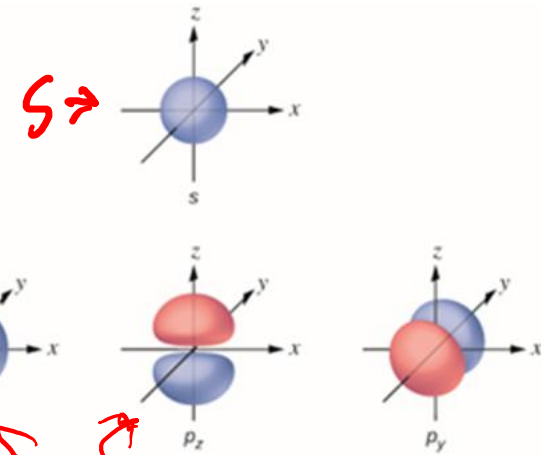
Solves bond angle issue

- Recall shapes and orientations of atomic orbitals →

- Consider oxygen's electron configuration



Orbital Diagram

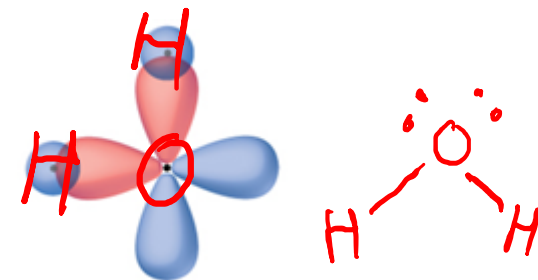


O uses

Draw in H_2O
Lewis Structure

- For H_2O , VBT would predict 90° between bonds, since 2 orthogonal p orbitals used

- Experimental evidence shows bond angle $\approx 104.5^\circ$ ($\ll 109.5^\circ$)
 - Recall: **bent** molecular geometry, w/in **tetrahedral** e^- pair geometry



Other Central Atom issues:

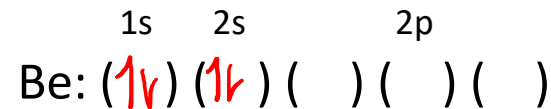
Be (BeF₂), B (BH₃), & C (CH₄)

Periodic Table of the Elements

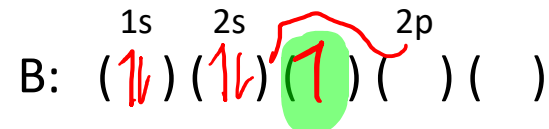
Period	1	2	13	14	15	16	17	18
Group	1	2	13	14	15	16	17	18
1	1 H 1.008 hydrogen							2 He 4.003 helium
2	3 Li 6.94 lithium	4 Be 9.012 beryllium	5 B 10.81 boron	6 C 12.01 carbon	7 N 14.01 nitrogen	8 O 16.00 oxygen	9 F 19.00 fluorine	10 Ne 20.18 neon

- Valence Bond theory predicts that a covalent bond requires an unpaired electron (from each atom)
- We know the above atoms are in Lewis structures, but

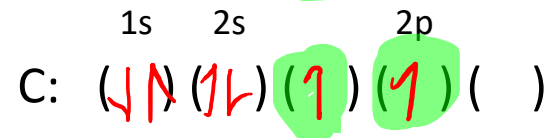
- A lone beryllium atom



- A lone boron atom



- A lone carbon atom



- Beryllium could form no bonds, boron one, and carbon only two bonds!
- VBT is therefore wrong both in orientation & # of bonds.

VBT Flawed; Remedy: Hybridization

4 Atomic 4 Hybrid.
 n orbitals in, n orbitals out

- **Hybridization**, whereby atomic orbitals are combined and then divided into equally shaped but differently oriented **hybrid orbitals**.

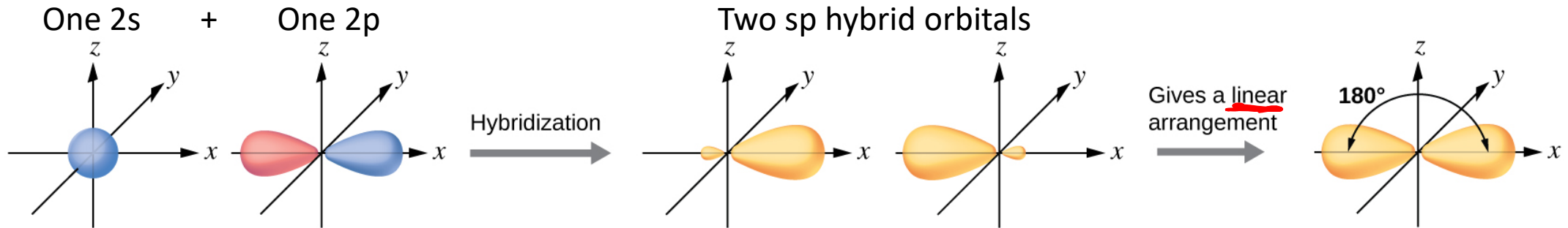
- For n atomic orbitals used, n hybrid orbitals are created.
- Shape & orientation of hybrid orbital $\neq$ atomic orbital
- All hybrid orbitals in a set are **Degenerate** (of **equivalent energy**).
- Hybrid orbital type for a particular atom depends on its **electron-pair geometry**.

Where
 A = central atom
 X = terminal atoms or lone pairs

Generic AX form
 (write in)

	180° AX_2	120° AX_3	109.5° CH_4	90° & 120° PCl_5	90° SF_6
e⁻ pair Geometry	Linear	Trigonal planar	Tetrahedral	Trigonal Bipyramidal	Octahedral
Hybrid Orbitals	sp	sp^2	sp^3	sp^3d	sp^3d^2

sp Hybrid Orbitals



- Consider BeF_2 - Be has two bonds

n orbitals in, n orbitals out

- Take atomic orbitals (A.O.s) of Be: $\begin{matrix} 1s & 2s & 2p & 2p & 2p \\ (1\downarrow) & (1\downarrow) & () & () & () \end{matrix}$

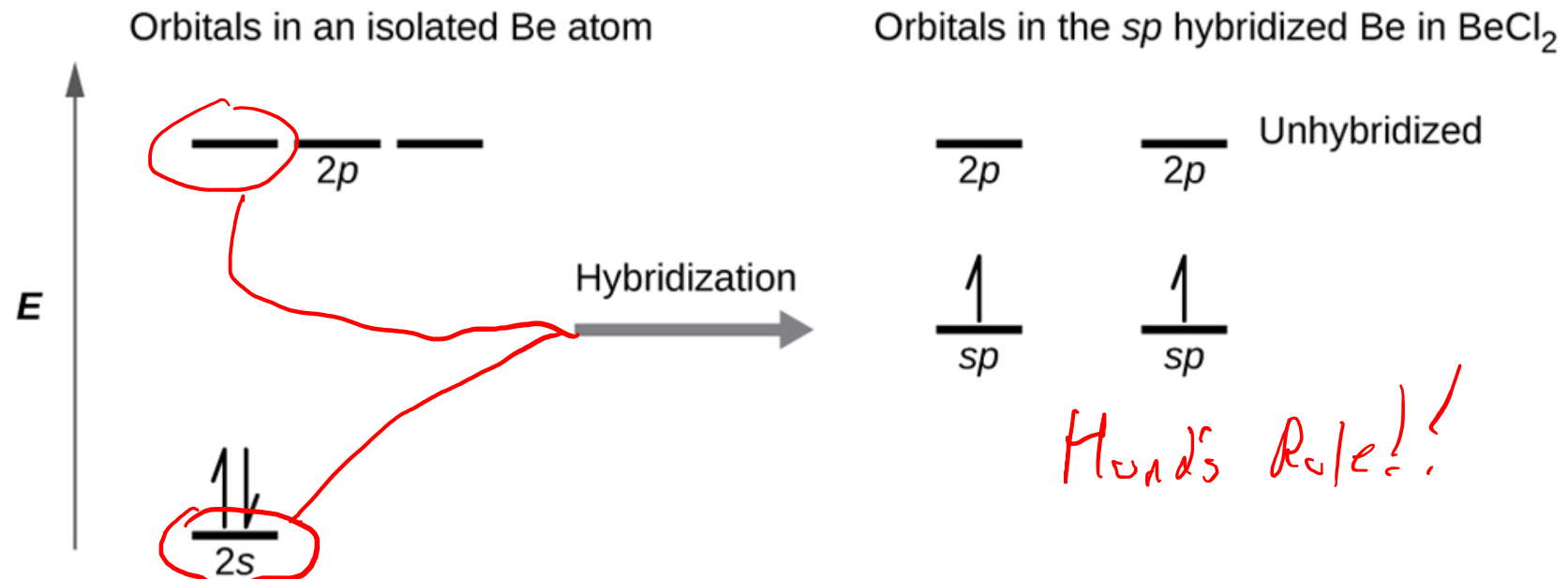
- If the 2s orbital is mixed with a single 2p orbital, two sp hybrid orbitals are created
 - Two orbitals in, two orbitals out.

- Then hybrid orbitals (H.O.s) of Be: $\begin{matrix} 1s & sp & sp & 2p & 2p \\ (1\downarrow) & (1) & (1) & () & () \end{matrix}$

- The two sp hybrid orbitals are arranged in a **linear** orientation

All hybrid orbitals in a set are **Degenerate** (of equivalent energy).

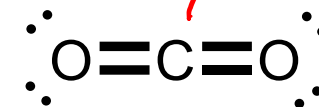
$s + p$
2 orbitals in, 2 orbitals out
 $2sp^2$



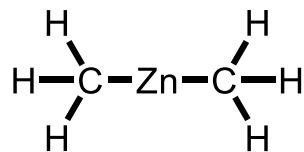
sp Hybridization

- sp hybridization is for any *central atom* with 2 groups
 - Recall: a group is a bound atom or a lone pair

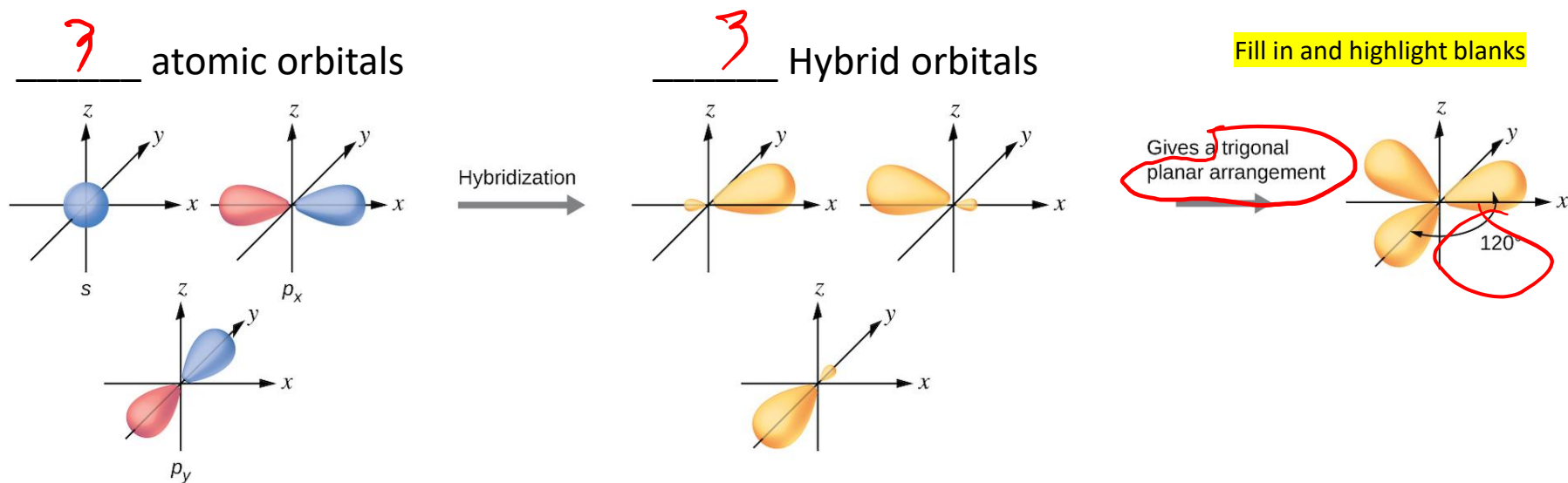
- Examples: HgCl_2 , $\text{Zn}(\text{CH}_3)_2$, HCCH , and CO_2 .



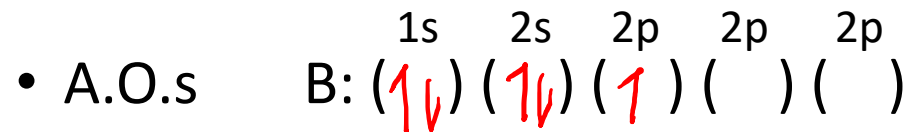
- Most common sp hybridization example is a C or N in a triple bond.



sp² Hybrid Orbitals

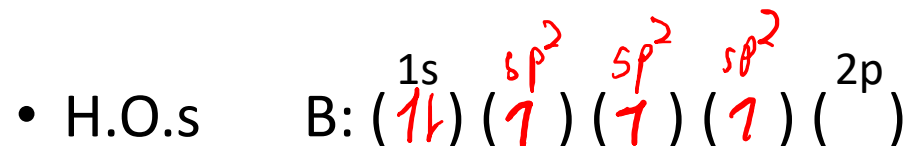


- Consider BH₃ - B has three bonds

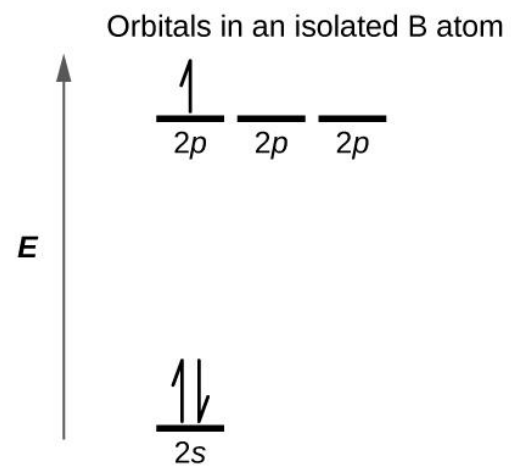
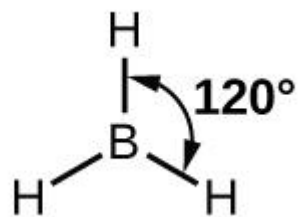


- If the 2s orbital is mixed with two 2p orbitals, three sp² hybrid orbitals are created
 - three orbitals in, three orbitals out

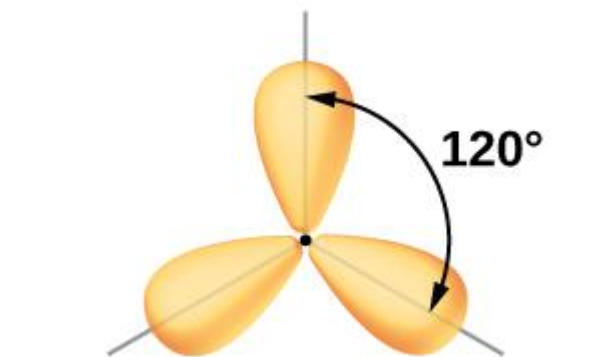
n orbitals in, n orbitals out



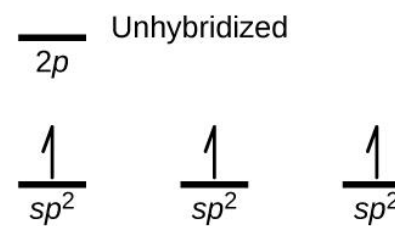
- The three sp² hybrid orbitals are arranged in a **trigonal planar** orientation



Hybridization
→



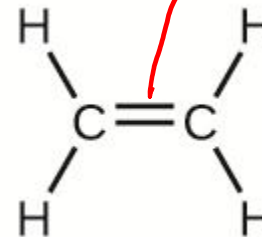
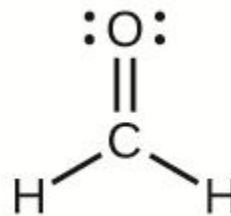
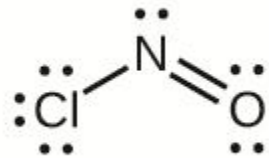
Orbitals in the sp^2 hybridized B atom in BH₃



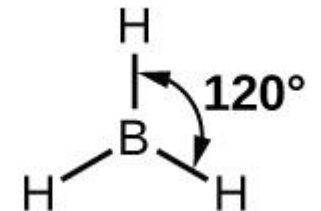
sp^2 Hybridization

sp^2

- sp^2 hybridization is for any *central atom with 3 groups*
 - Recall: a group is a bound atom or a lone pair
- Examples: ClNO , CH_2O , H_2CCH_2 .
 - Most common example is a C, N, or O in a double bond.



σ & a π bond
Bond.



sp^3 Hybrid Orbitals

- Consider CH_4 - C has four bonds

Take C: $\overset{1s}{(1\downarrow)} \overset{2s}{(1\downarrow)} \overset{2p}{(1)} \overset{2p}{(1)} \overset{2p}{()}$

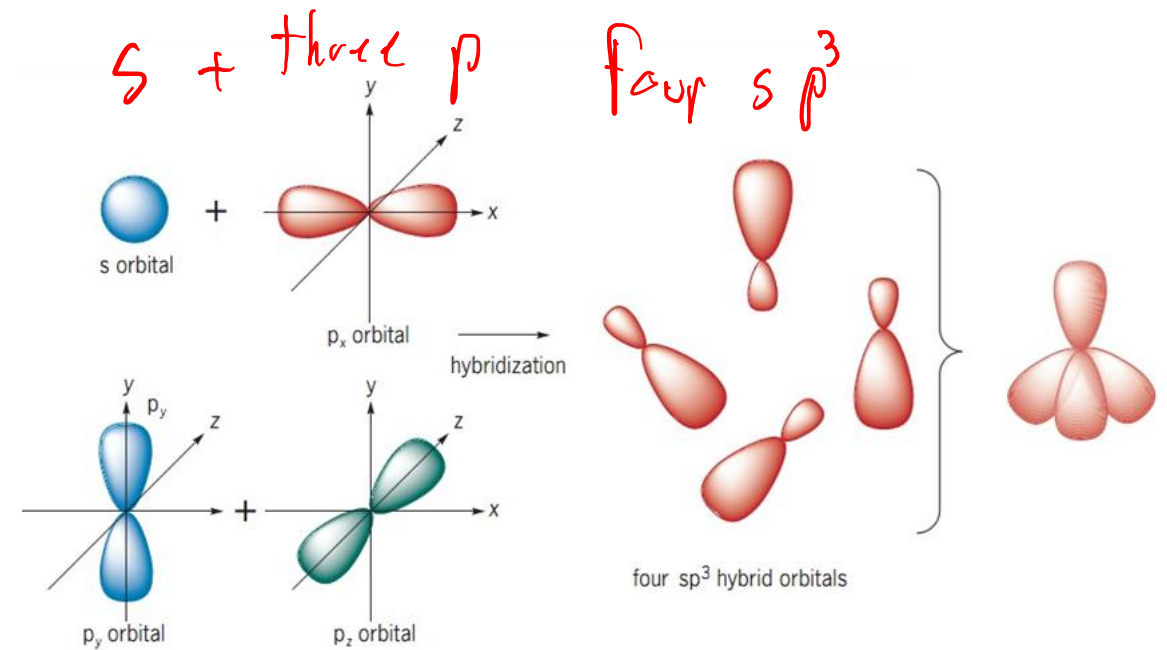
- If the 2s orbital is mixed with three 2p orbitals, four sp^3 hybrid orbitals are created

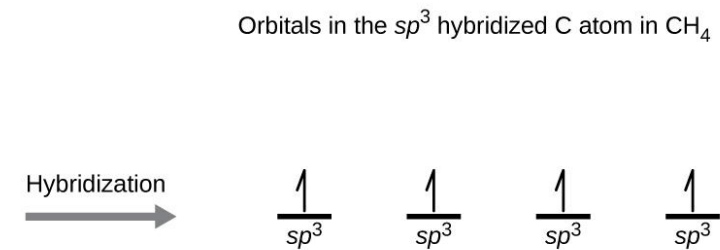
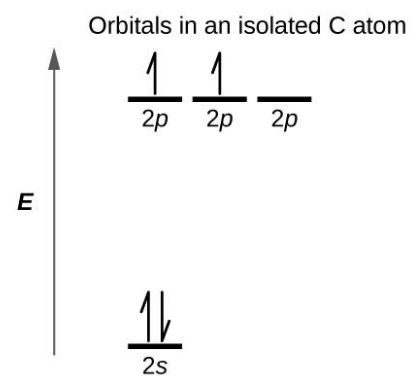
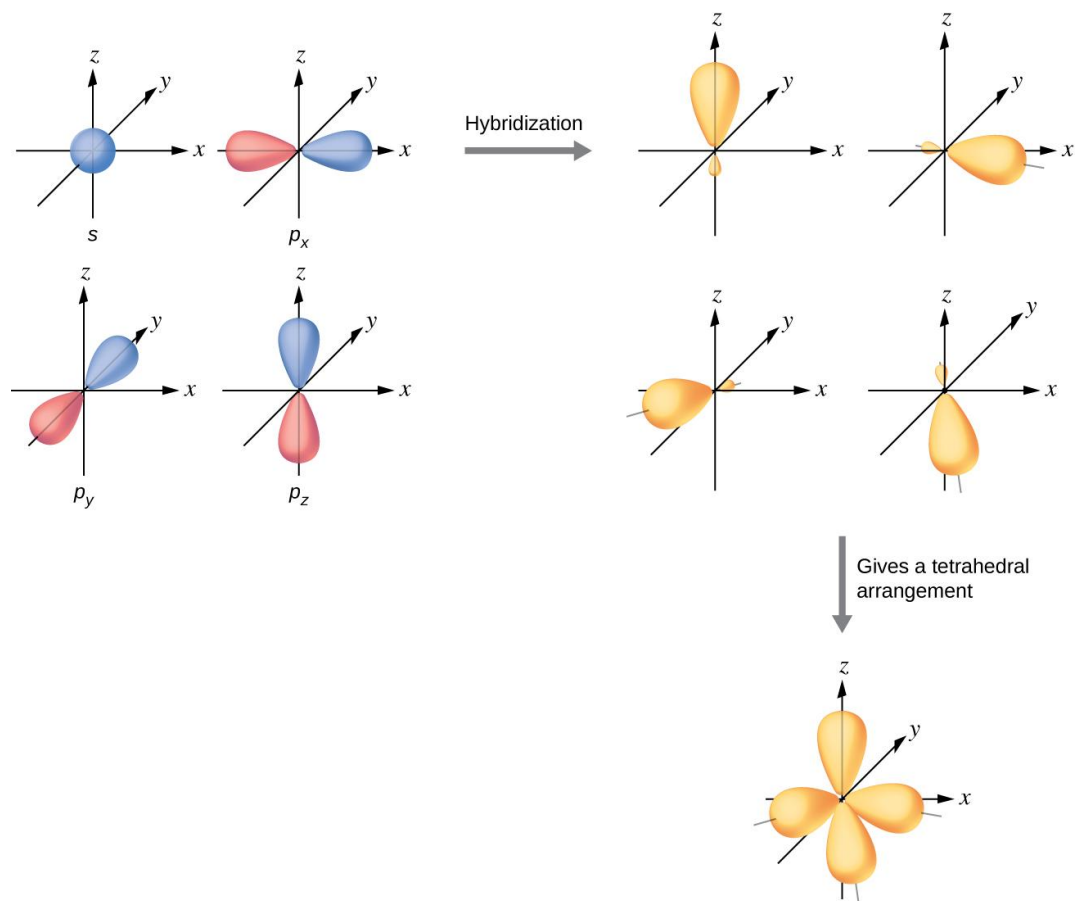
- Four orbitals in, four orbitals out

n orbitals in, n orbitals out

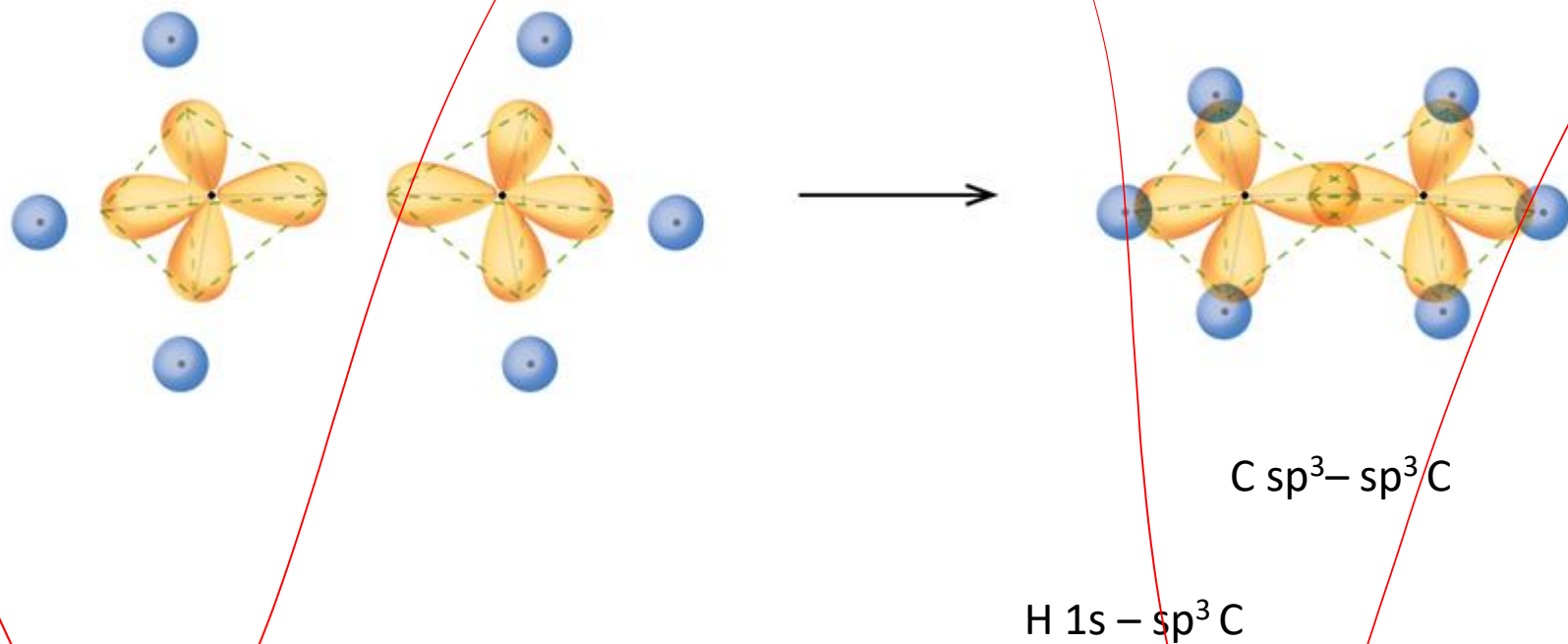
- sp^3 hybrid orbitals arranged in a **tetrahedral** orientation

Then C: $\overset{1s}{(1\downarrow)} \overset{sp^3}{(1)} \overset{sp^3}{(1)} \overset{sp^3}{(1)} \overset{sp^3}{(1)}$

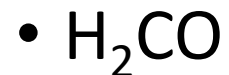




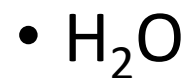
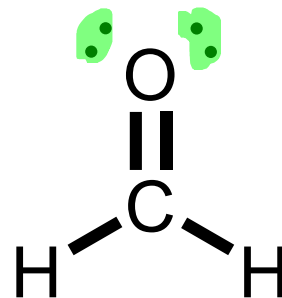
Orbitals in Ethane (C_2H_6)



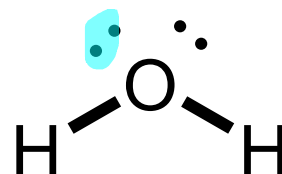
Hybrid orbitals can hold a lone pair of e⁻



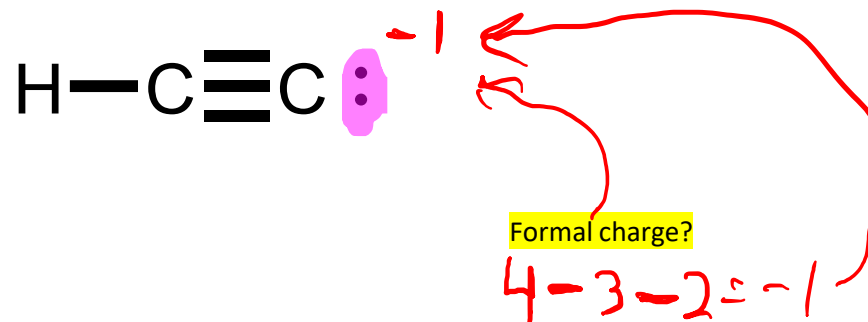
- L.p. in sp² hybrid orbital



- L.p. in sp³ hybrid orbital



- L.p. in sp hybrid orbital



ICE Hybrid Orbitals

- 2 mins
- Octet Rule?

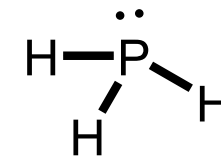
Period	Group
1	1
1	1
2	3
2	4

1	H 1.008 hydrogen
3	Li 6.94 lithium
4	Be 9.012 beryllium

Periodic Table of the Elements

							18
					2		He 4.003 helium
13	14	15	16	17			
5 B 10.81 boron	6 C 12.01 carbon	7 N 14.01 nitrogen	8 O 16.00 oxygen	9 F 19.00 fluorine	10 Ne 20.18 neon		

sp³d and sp³d² Hybrid Orbitals



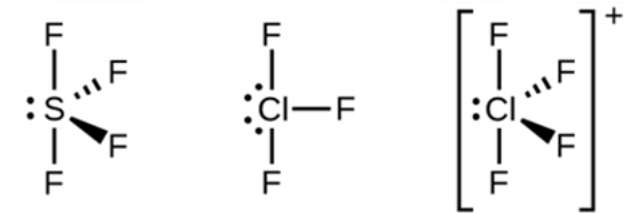
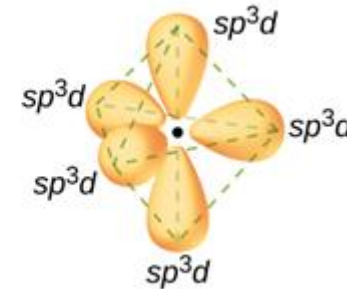
May not

PH₃ – sp³ hybridization

- Atoms that **may** make use of expanded octets use d atomic orbitals to form hybrid orbitals

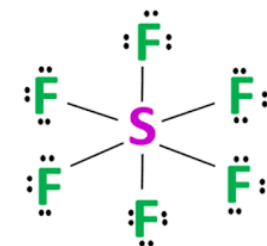
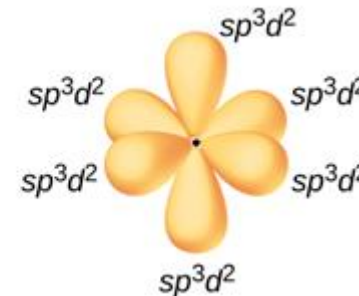
- 5 sp³d hybrid orbitals from an s, three p and a d atomic orbitals.

- Ex: SF₄, ClF₃



- 6 sp³d² hybrid orbitals from an s, three p and two d atomic orbitals.

- Ex: SF₆, XeF₆



Assigning Hybrid Orbitals to Central Atoms

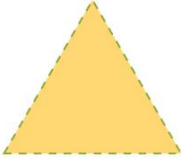
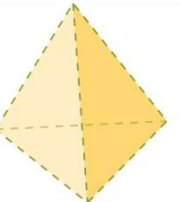
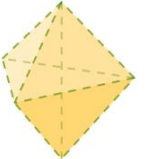
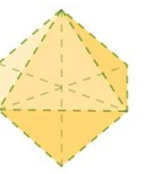
Black dot is the nucleus of the atom w/ hybrid orbitals

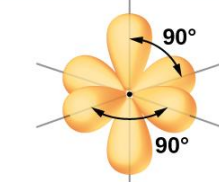
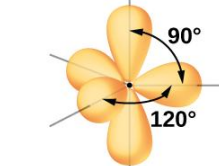
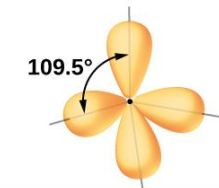
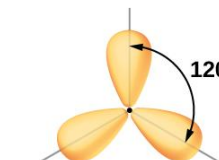
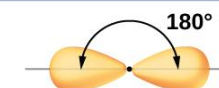
- Solely dependent on **number of groups**.

- Group = a bound atom or a lone pair of electrons.

- Note: electron-pair geometries column

Generic AX form
(write in)

Regions of Electron Density	Arrangement	e ⁻ pair Geometry	Hybridization
2		linear	<i>sp</i>
3		trigonal planar	<i>sp</i> ²
4		tetrahedral	<i>sp</i> ³
5		trigonal bipyramidal	<i>sp</i> ³ <i>d</i>
6		<u>octahedral</u>	<i>sp</i> ³ <i>d</i> ²



Example Compound
HCCH

H₂CCH₂
BF₃

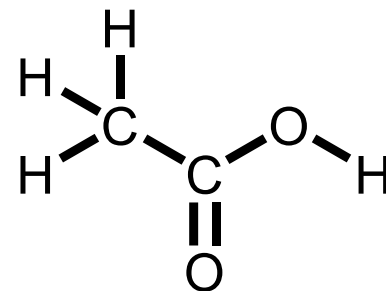
CH₄
H₂O
NH₃

SF₄
PCl₅

SF₆

Practice

- What is the hybridization of S in SO_4^{2-} and Se in SeF_4 and the two C's in acetic acid (given structure)?
 - Requires drawing Lewis structures.



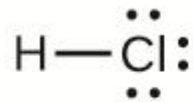
Periodic Table of the Elements

Period		Group																18			
1		1	2													13	14	15	16	17	2
1	H															B	C	N	O	F	He
	1.008															10.81	12.01	14.01	16.00	19.00	4.003
	hydrogen															boron	carbon	nitrogen	oxygen	fluorine	helium
2	Li	Be														5	6	7	8	9	10
	6.94	9.012														B	C	N	O	F	Ne
	lithium	beryllium														10.81	12.01	14.01	16.00	19.00	20.18
3	Na	Mg														13	14	15	16	17	18
	22.99	24.31														Al	Si	P	S	Cl	Ar
	sodium	magnesium														26.98	28.09	30.97	32.06	35.45	39.95
																aluminum	silicon	phosphorus	sulfur	chlorine	argon
			3	4	5	6	7	8	9	10	11	12									

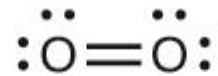
Multiple bonds: Refresher

single bond = σ bond.

- A double bond is composed of 1 σ and 1 π bond
 - $\text{C}=\text{C}$
- A triple bond is composed of 1 σ and 2 π bond
 - $\text{C}\equiv\text{C}$



One σ bond
No π bonds



One σ bond
One π bond



One σ bond
Two π bonds

1127 Housekeeping Oct 14th Exam II material

To the end of this chapter

- Quiz 4 – Done
- Exam II – Thurs Oct 16th – covers Quiz 3 & 4 and Chapter 5 (Oct 9th & first part of today).
- HW see due dates →
 - Aktiv Quiz Daily Feedback
 - MIDTERM GRADES
- Extra Credit NACE Journal 3

Due tonight!!

A6: Chapter 4 part 2: Lewis Structures & Formal Charges

Homework

Start Date: Monday, Sep 29, 2025 1:28 PM

Due Date: Tuesday, Oct 14, 2025 11:59 PM

100 problems

DUE IN 7 DAYS

A7: Chapter 4 part 3: electron pair/molecular Geometries, Bond Angles, Polar Molecules, Intermolecular Forces, Solubility

Homework

Start Date: Monday, Sep 29, 2025 1:32 PM

Due Date: Thursday, Oct 16, 2025 1:59 AM

70 problems

DUE IN 13 DAYS

A8: Chapter 5 Advanced Theories of Bonding

Homework

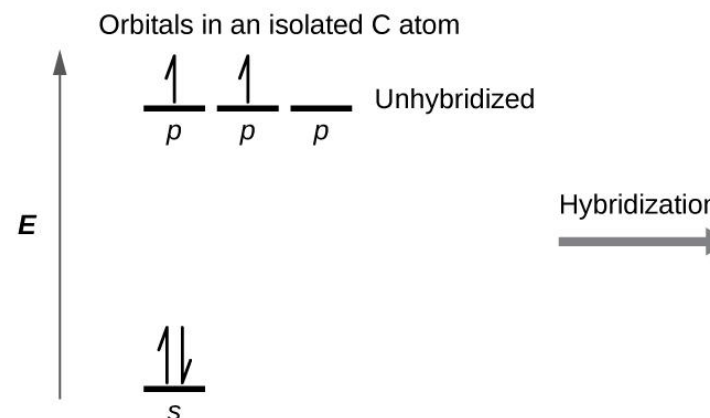
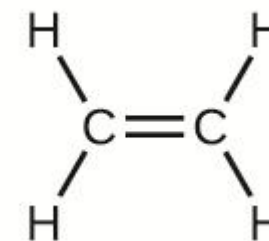
Start Date: Monday, Feb 26, 2024 1:38 PM

Due Date: Tuesday, Oct 21, 2025 11:59 PM

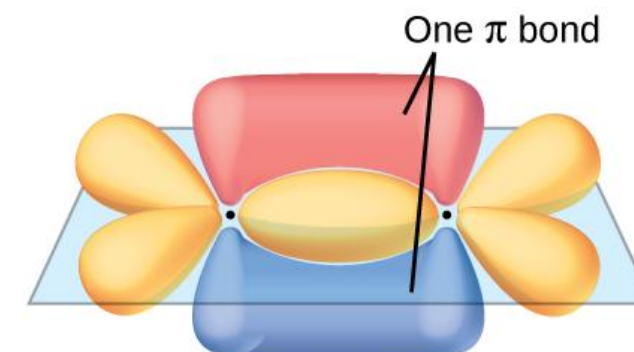
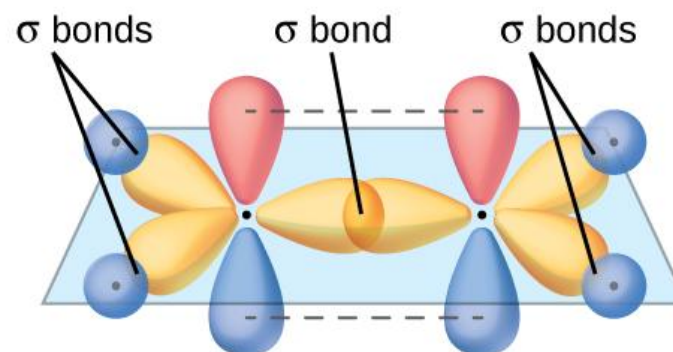
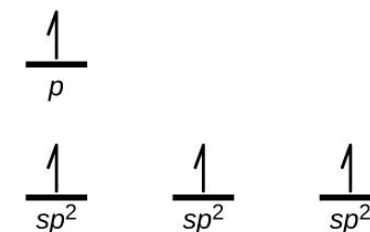
40 problems

Double Bonds – Ethylene (C_2H_4)

- **Pi π bond** is formed from side-by-side **overlap** of **unhybridized p orbitals**
- All 6 atoms (dots in below picture) in the same plane
 - Pi bond above & below the plane



Orbitals in the sp^2 hybridized C atom in C_2H_4



ICE Planarity of Molecules

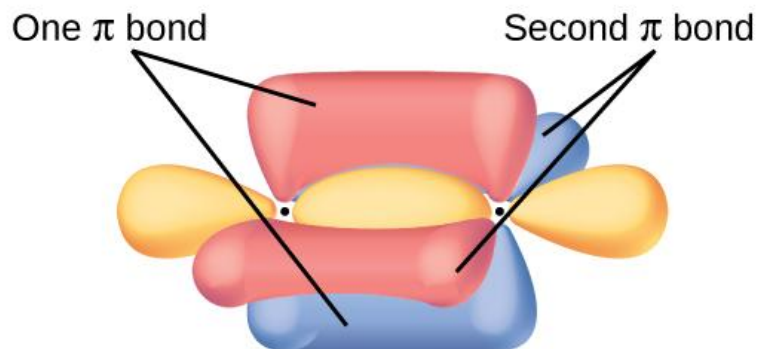
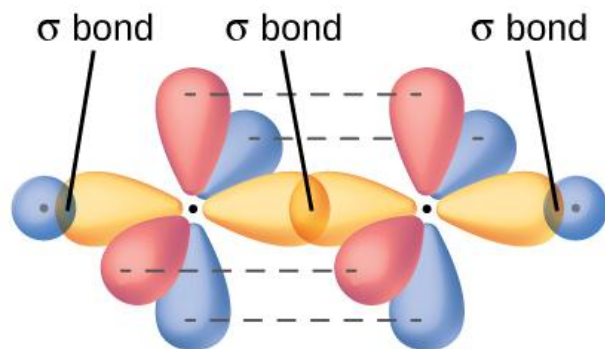
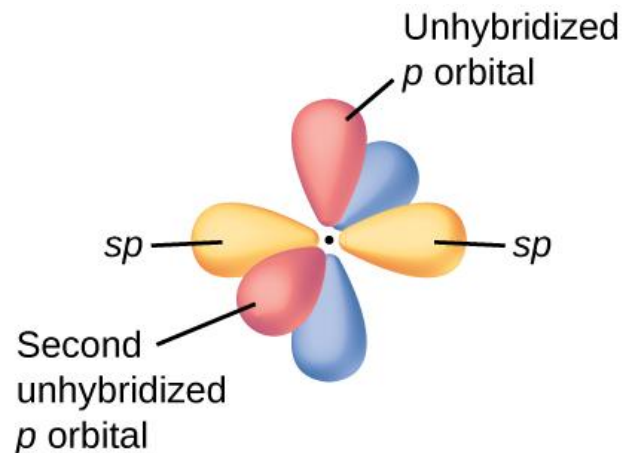
- 30 s

trigonal planar $\approx sp^2$

Triple Bonds – Acetylene (C_2H_2)



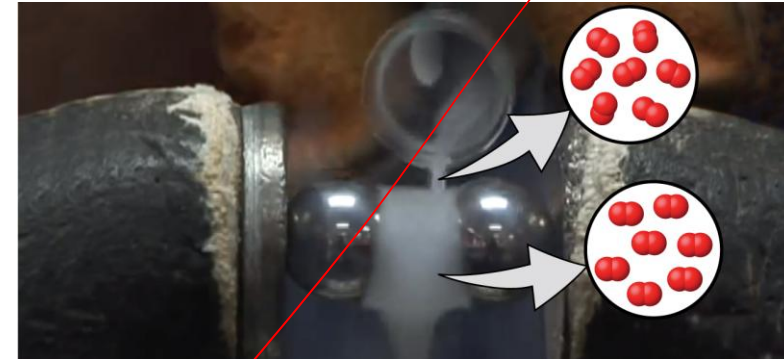
- Two pi bonds are formed from side-by-side overlap of **unhybridized** p orbitals
 - Pi bonds are **orthogonal** (90° apart)



Hybridization involves only σ bonds, ~~lone pairs~~ of electrons, and single unpaired electrons (radicals).

Molecular Orbital Theory

- O_2 exhibits **paramagnetism** where it is attracted to a magnetic field.
 - Paramagnetism comes from **unpaired electrons**.
 - Experiments show that O_2 has 2 unpaired electrons.
 - But in Lewis structure for O_2 , all e^- are paired.
 - **Diamagnetism** – when all electrons are paired, such species *weakly repel* a magnetic field.
- **Solution:** Molecular Orbital Theory

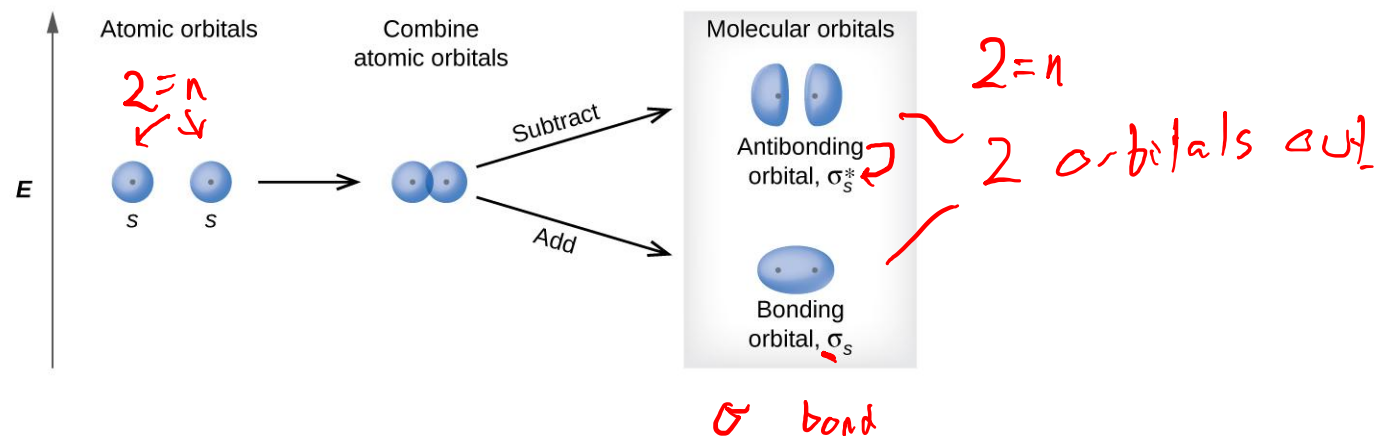


Comparison of Bonding Theories

Valence Bond Theory	Molecular Orbital Theory
considers bonds as localized between one pair of atoms	considers electrons delocalized throughout the entire molecule
creates bonds from overlap of atomic orbitals (s , p , $d...$) and hybrid orbitals (sp , sp^2 , $sp^3...$)	combines atomic orbitals to form molecular orbitals (σ , σ^* , π , π^*)
forms σ or π bonds	creates bonding and antibonding interactions based on which orbitals are filled
predicts molecular shape based on the number of regions of electron density	predicts the arrangement of electrons in molecules
needs multiple structures to describe resonance	

Molecular Orbital Theory

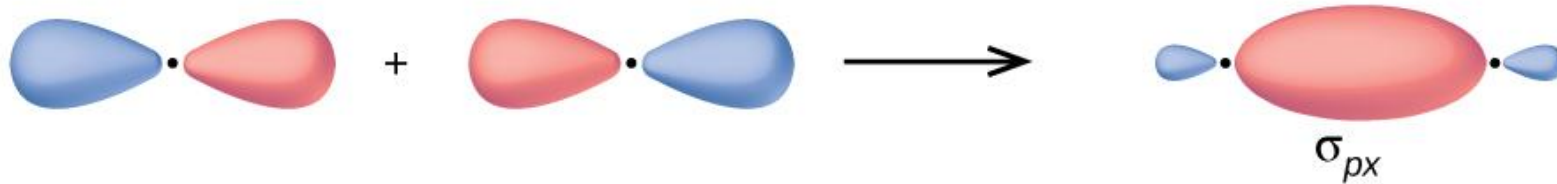
Still, ***n orbitals in, n orbitals out***



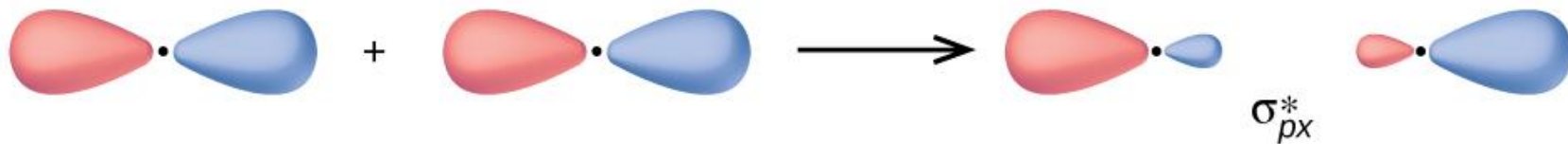
Sigma bonds in Molecular Orbital Theory

Draw in arrows of atoms approaching

Same amplitude (colors) overlap, **bonding orbital** occurs

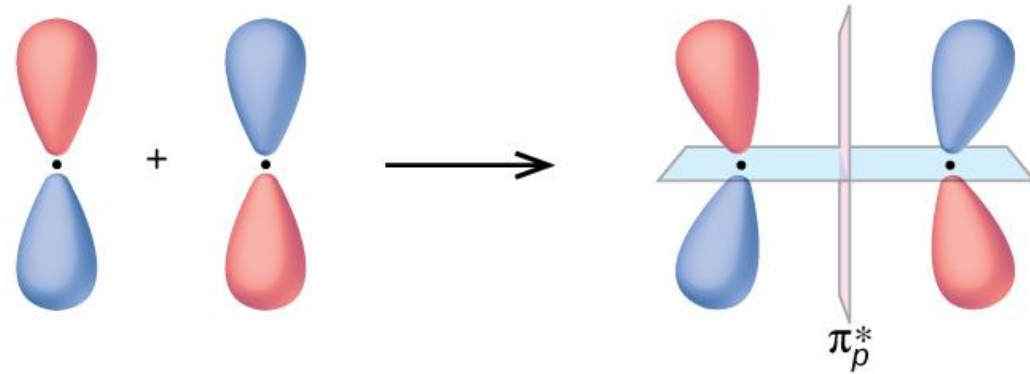


Opposite amplitudes (colors) overlap, **antibonding (indicated by *) orbital** occurs

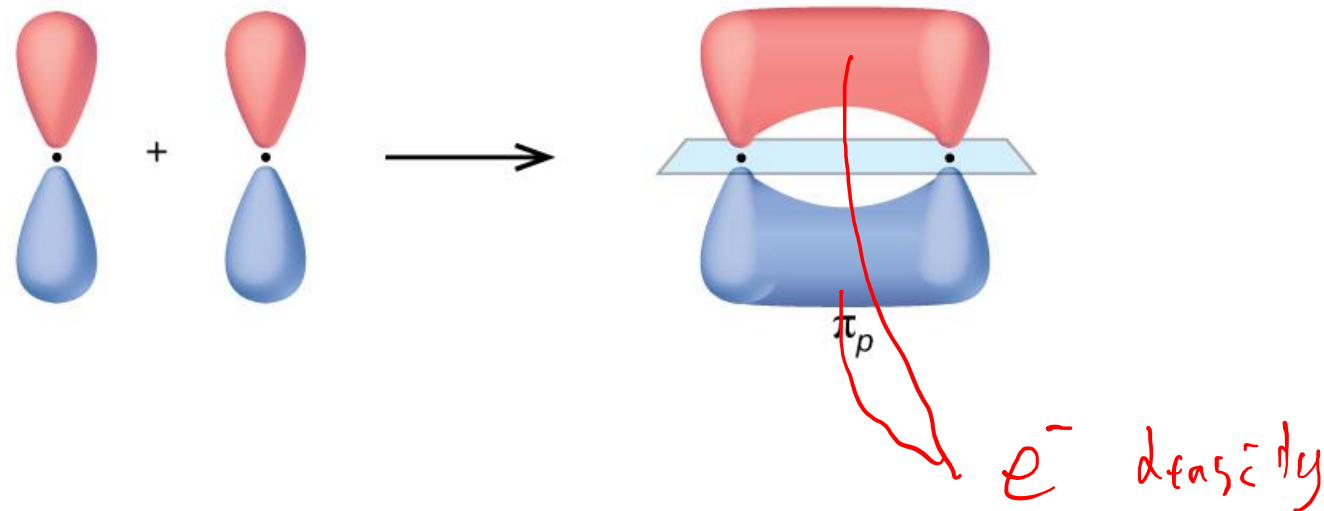


Pi bonds in Molecular Orbital Theory

Opposite amplitudes (colors) overlap, antibonding pi orbital (π^*) occurs

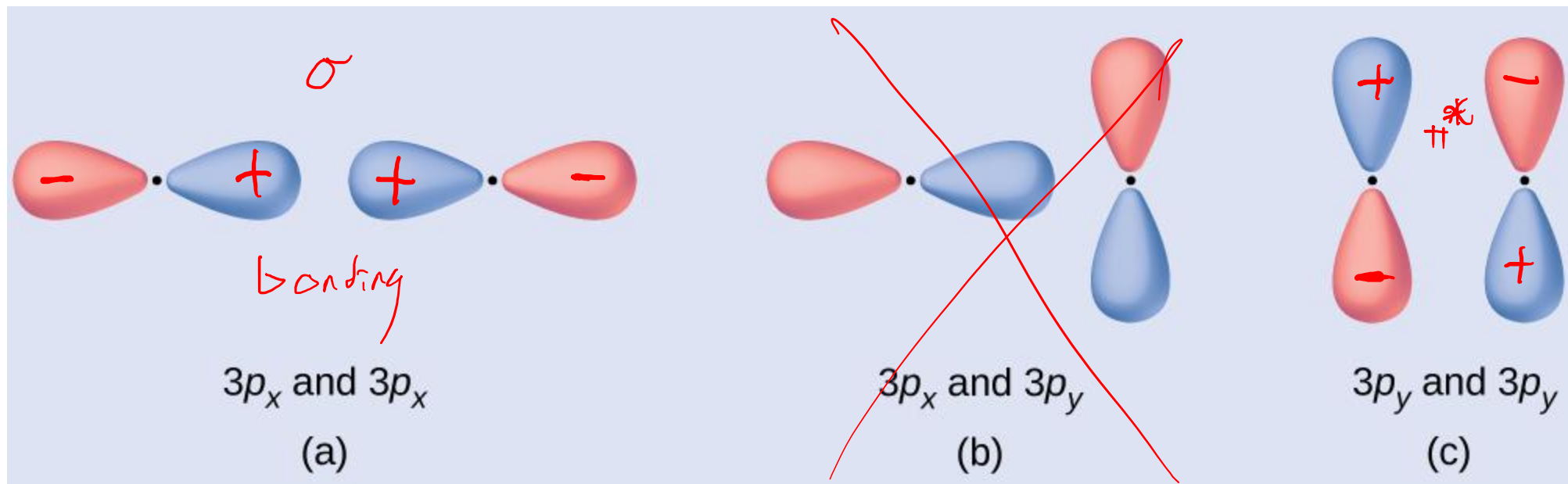


Same amplitude (colors) overlap, bonding pi orbital (π) occurs



Practice: σ σ^* π π^*

- Predict what type (if any) of molecular orbital would result from adding the wave functions so each pair of orbitals shown overlap. The orbitals are all similar in energy.



Periodic Table of the Elements

Period	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 H 1.008 hydrogen																	2 He 4.003 helium
2	3 Li 6.94 lithium	4 Be 9.012 beryllium											5 B 10.81 boron	6 C 12.01 carbon	7 N 14.01 nitrogen	8 O 16.00 oxygen	9 F 19.00 fluorine	10 Ne 20.18 neon
3	11 Na 22.99 sodium	12 Mg 24.31 magnesium											13 Al 26.98 aluminum	14 Si 28.09 silicon	15 P 30.97 phosphorus	16 S 32.06 sulfur	17 Cl 35.45 chlorine	18 Ar 39.95 argon
4	19 K 39.10 potassium	20 Ca 40.08 calcium	21 Sc 44.96 scandium	22 Ti 47.87 titanium	23 V 50.94 vanadium	24 Cr 52.00 chromium	25 Mn 54.94 manganese	26 Fe 55.85 iron	27 Co 58.93 cobalt	28 Ni 58.69 nickel	29 Cu 63.55 copper	30 Zn 65.38 zinc	31 Ga 69.72 gallium	32 Ge 72.63 germanium	33 As 74.92 arsenic	34 Se 78.97 selenium	35 Br 79.90 bromine	36 Kr 83.80 krypton
5	37 Rb 85.47 rubidium	38 Sr 87.62 strontium	39 Y 88.91 yttrium	40 Zr 91.22 zirconium	41 Nb 92.91 niobium	42 Mo 95.95 molybdenum	43 Tc [97] technetium	44 Ru 101.1 ruthenium	45 Rh 102.9 rhodium	46 Pd 106.4 palladium	47 Ag 107.9 silver	48 Cd 112.4 cadmium	49 In 114.8 indium	50 Sn 118.7 tin	51 Sb 121.8 antimony	52 Te 127.6 tellurium	53 I 126.9 iodine	54 Xe 131.3 xenon
6	55 Cs 132.9 cesium	56 Ba 137.3 barium	57-71 La-Lu *	72 Hf 178.5 hafnium	73 Ta 180.9 tantalum	74 W 183.8 tungsten	75 Re 186.2 rhenium	76 Os 190.2 osmium	77 Ir 192.2 iridium	78 Pt 195.1 platinum	79 Au 197.0 gold	80 Hg 200.6 mercury	81 Tl 204.4 thallium	82 Pb 207.2 lead	83 Bi 209.0 bismuth	84 Po [209] polonium	85 At [210] astatine	86 Rn [222] radon
7	87 Fr [223] francium	88 Ra [226] radium	89-103 Ac-Lr **	104 Rf [267] rutherfordium	105 Db [270] dubnium	106 Sg [271] seaborgium	107 Bh [277] bohrium	108 Hs [277] hassium	109 Mt [276] meitnerium	110 Ds [281] darmstadtium	111 Rg [282] roentgenium	112 Cn [285] copernicium	113 Nh [285] nihonium	114 Fl [289] flerovium	115 Mc [288] moscovium	116 Lv [293] livermorium	117 Ts [294] tennessine	118 Og [294] oganesson

57 La 138.9 lanthanum	58 Ce 140.1 cerium	59 Pr 140.9 praseodymium	60 Nd 144.2 neodymium	61 Pm [145] promethium	62 Sm 150.4 samarium	63 Eu 152.0 europium	64 Gd 157.3 gadolinium	65 Tb 158.9 terbium	66 Dy 162.5 dysprosium	67 Ho 164.9 holmium	68 Er 167.3 erbium	69 Tm 168.9 thulium	70 Yb 173.1 ytterbium	71 Lu 175.0 lutetium
89 Ac [227] actinium	90 Th 232.0 thorium	91 Pa 231.0 protactinium	92 U 238.0 uranium	93 Np [237] neptunium	94 Pu [244] plutonium	95 Am [243] americium	96 Cm [247] curium	97 Bk [247] berkelium	98 Cf [251] californium	99 Es [252] einsteinium	100 Fm [257] fermium	101 Md [258] mendelevium	102 No [259] nobelium	103 Lr [262] lawrencium

Diagram illustrating the components of an element box (Hydrogen):

- Atomic number: 1
- Symbol: **H**
- Atomic mass: 1.008
- Name: hydrogen

Color Code	
 Metal	Solid
 Metalloid	Liquid
 Nonmetal	Gas